

Link to the dataset: [link](#)

About the dataset

The dataset consists of conversational text collected from real messaging exchanges. The text is rich in codemixed instances across multiple languages. A sample is attached. Each row represents one conversation; the relevant columns for your task are `final_message_list` (individual messages per turn) and `final_speaker_tracking` (the corresponding speaker label per turn).

Your task

This task has three components. Please read all of them before starting.

Component 1: Language identification annotation

Your primary annotation task is word-level language identification. For each token in a message, the label should indicate the language it belongs to. At minimum, consider the following categories: English, Hokkien, Malay, Tamil, and Other (non-linguistic tokens such as emoji, URLs, or untranscribable content).

You are free to refine or expand this tagset if you believe it would be more useful for downstream tasks. If you do so, please justify your choices. There is no single correct tagset here; we are evaluating how you think about the problem.

Component 2: Hate speech and harmful content labelling

Design and apply a labelling scheme for hate speech and harmful content. Define what you mean by hate speech in the context of codemixed conversational data. Your definition should account for the fact that offensive or derogatory language may be expressed through non-English terms that would not be flagged by an English-only classifier.

You are free to determine the granularity and structure of this scheme. Consider whether binary labelling is sufficient or whether you need finer categories.

Component 3: LLM-based annotation pipeline

Given the volume of data, human annotation of every sample is not feasible. You should therefore design an LLM-assisted annotation approach and provide working code for two scenarios:

Scenario A (API-based): Write code that calls an external model API (OpenAI or Anthropic, your choice) to perform language identification and/or hate speech labelling on a given input message. You do not need to run this code or incur API costs; a clean, runnable implementation with a clear prompt and structured output is sufficient.

Scenario B (open-source models): Write code that uses a locally runnable open-source model to perform the same annotation task. Run this on a small number of examples from

the provided dataset (5 to 10 turns is enough) and include the model outputs. Small models are entirely acceptable; the goal is to demonstrate that the pipeline works end-to-end.

Human annotation guidelines

As part of your submission, write a brief annotation guideline document intended for human annotators. This should cover: the annotation task definition, the label categories and their decision criteria, examples of ambiguous cases and how to handle them, and any quality control expectations. We use a structure of one primary annotator and two secondary annotators, as you described in your interview, so the guidelines should be written with that workflow in mind.

Submission

Please prepare a written report covering the following:

1. A brief description of the dataset based on your own reading of the sample – what you observed and what challenges you anticipate for annotation.
2. Your annotation scheme design decisions for both tasks, including your rationale.
3. Your LLM pipeline design: prompt choices, model selection, and a reflection on the quality of outputs from your open-source run – where the model performed well, where it failed, and whether your prompt needed iteration.
4. Potential future directions – what additional annotation layers, model improvements, or research questions you think would be valuable.

In addition to the report, your submission should include your annotation outputs, code for both scenarios, and the human annotation guidelines document.

All files must be submitted via a shared Google Drive folder or a GitHub repository.

Optional extension: Disclosure annotation

If you have time and interest, as an optional component outline how you would approach text-based annotation of disclosure from the conversations: how you would define disclosure in this context, what dimensions you would annotate, and what an annotation scheme and associated human guidelines might look like. Code is not required but would be a bonus.